SWAYAM NPTEL COURSE ON

INDUSTRIAL WASTEWATER TREATMENT

Prof. Sunil Kumar Gupta

Department of Environmental Science & Engineering

IIT (ISM) Dhanbad, Jharkhand

Lecture 01: Adsorption Process

CONCEPTS COVERED

- Adsorption Phenomenon
- Concept & Classification
- Adsorption isotherms
- Batch and Column Studies
- Numerical

Kindly do not use this portion, because here Teacher front view will appear.

NPTEL
JAN 2024

Adsorption

- **Definition:** Adsorption can be defined as a process in which atoms or molecules move from a bulk phase (i.e., solid, liquid or gas) onto a solid or liquid surface.

or

- In other words, it is **adhesion of atoms**, ions or molecules from a gas liquid or dissolved solid to a surface.
- **Adsorbate** – The substance which **is adsorbed on the surface**.
- **Adsorbent** – The substance on which surface the adsorbate is adsorbed.
- Adsorption can occur between two phases, namely **liquid–liquid**, **gas–liquid**, **gas–solid**, or **liquid–solid** interfaces.
- The process for adsorbates to release from adsorbent surfaces into water – **Desorption.**
- When activated carbon is used, the adsorbing phase is a solid.

Absorption Vs Adsorption

Particulars	Absorption	Adsorption
Phenomenon	It is a bulk phenomenon	It is a surface phenomenon.
Definition	Assimilation of molecular species throughout the bulk of the solid or liquid is termed as absorption.	Accumulation of the molecular species at the surface rather than in the bulk of the solid or liquid is termed as adsorption.
Heat exchange	Endothermic process	Exothermic process
Temperature	It is **not affected** by temperature	It is **favoured** at **low temperature**
Rate of reaction	It occurs at a uniform rate.	It steadily increases and reach to equilibrium
Concentration	It is same throughout the material.	Concentration gradient: Concentration on the surface of adsorbent is different from that in the bulk.

NPTEL
JAN 2024

There are two types of adsorption: Physisorption and Chemisorption

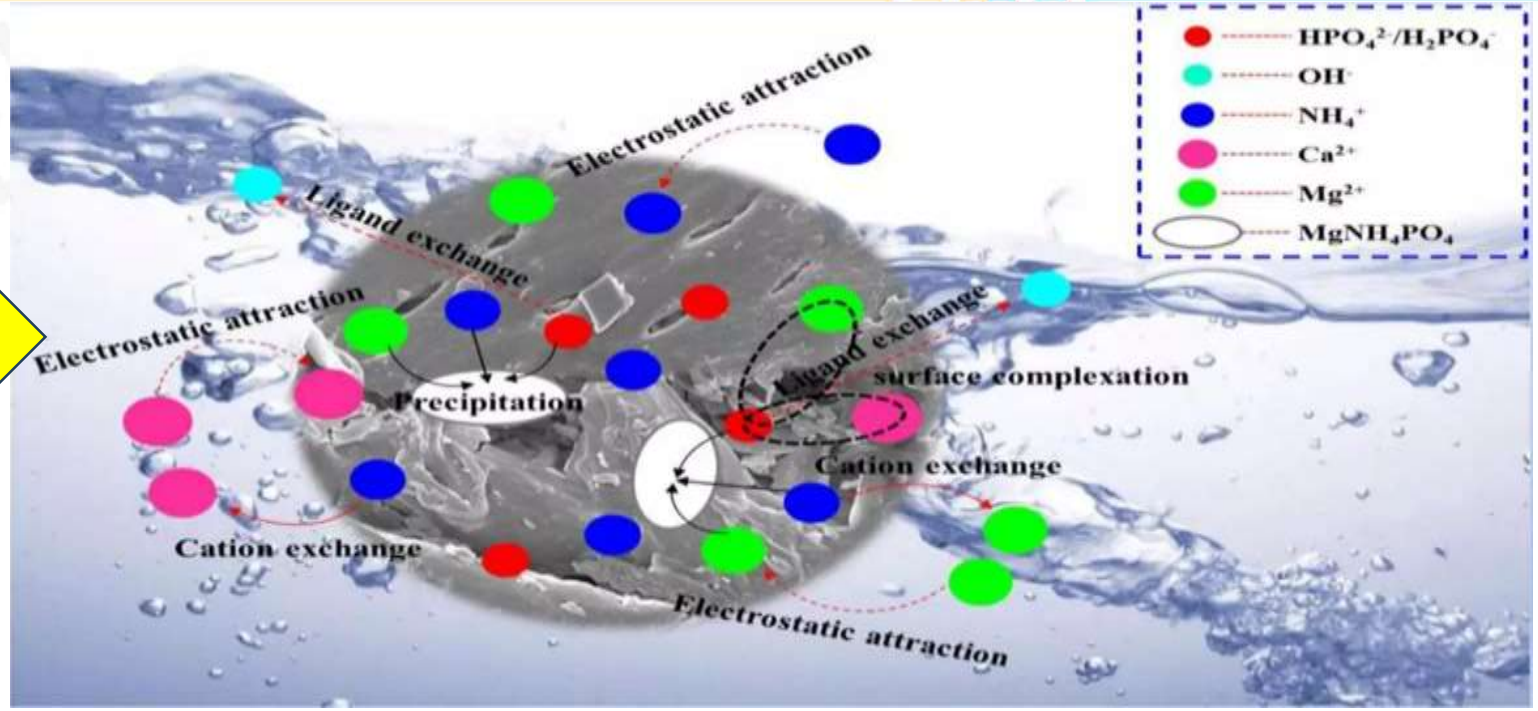
Physisorption	Chemisorption
It arises because of van der Waals forces.	It is caused by chemical bond formation.
It is not specific in nature.	It is highly specific in nature.
It is reversible in nature.	It is irreversible in nature.
It depends on the nature of gas. More easily liquefiable gases are adsorbed readily.	It also depends on the nature of gas. Gases which can react with adsorbent show chemisorption.
Enthalpy of adsorption is low (20-40 k J mol^{-1})	Enthalpy of adsorption is high (80-240 k J mol^{-1}).
Favourable with low temperature. It decreases with increase in temperature.	Favourable at high temperature. It increases with increase in temperature.
No appreciable activation energy is needed	High activation energy is sometimes needed.
It increases with increase of the surface area of adsorbent.	It too increases with an increase of surface area of adsorbent.
It results into multi molecular layers on adsorbent surface under high pressure.	It results into uni-molecular layer on adsorbent surface.

NPTEL
JAN 2024

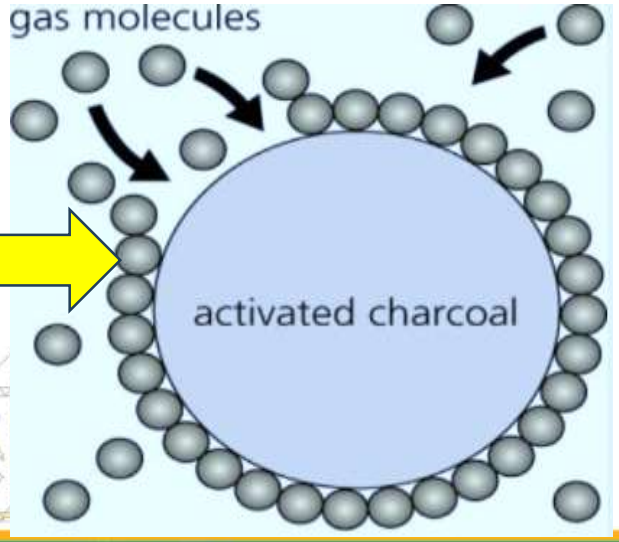
Adsorption Principle

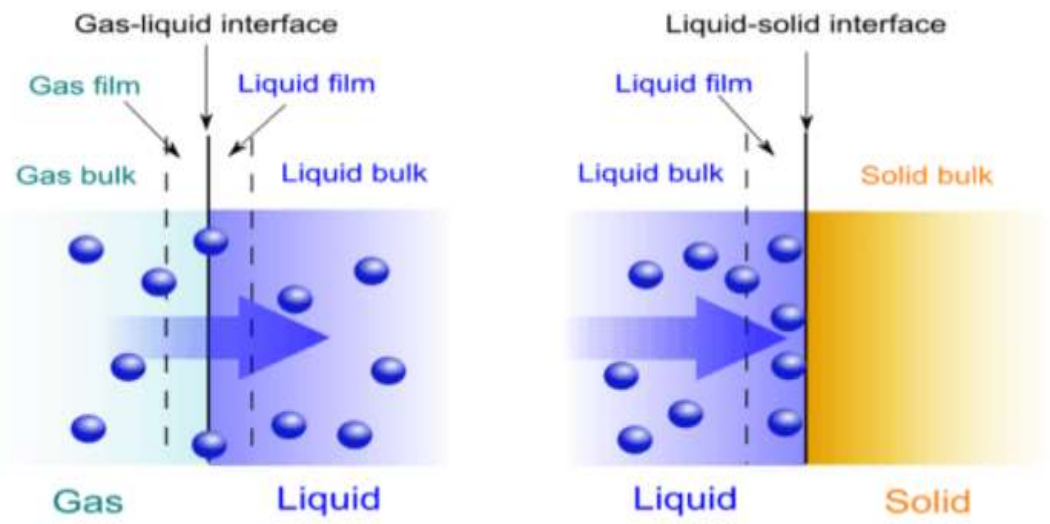
- Adsorption is a surface phenomenon & consequence of surface energy.
- Surface of liquid or solid is in a state of strain which results in unbalanced residual force at the surface thereby resulting higher surface energy.
- The surface of liquids/solids always have a tendency to attract & retain the molecular species with which it comes in contact.

Adsorption Mechanism



Adsorption Process





Types of Commercial Adsorbents

- Activated Carbon
- Silica Gel
- Activated Alumina
- Synthetic Zeolite
- Clay
- Polymers and Resins

- Synthetic **polymeric and silica-based adsorbents** are **seldom used because of their high cost.**
- **Activated carbon** is **used most frequently** as an advanced treatment for removal of hazardous/non-biodegradable impurities.

Types of Commercial Adsorbent material

Activated Carbon: Preparation and Properties

- Activated Carbon- Produced through molding, carbonization and activation of coal or wood material; lignite, peat, nut shell, coconut shell, lignin, petroleum coke and synthetic high polymers.
- Category: Granular activated carbon (GAC) & Powdered activated carbon (PAC).

Note: **Steam and CO_2** gases develop a **porous like structure**, in the char and thus creates a large internal surface area 700-1200 m^2/g .

Microporous: pore dia< 4 nm

Selection of Base materials (nut shell, coconut shell, lignin etc.)

↓

Heated to red heat @ 700 °C (Insufficient Oxygen)

↓

Char as Product

↓

Adding steam and CO_2 gases at 800 to 900°C

↓

Activated Carbon (final product)

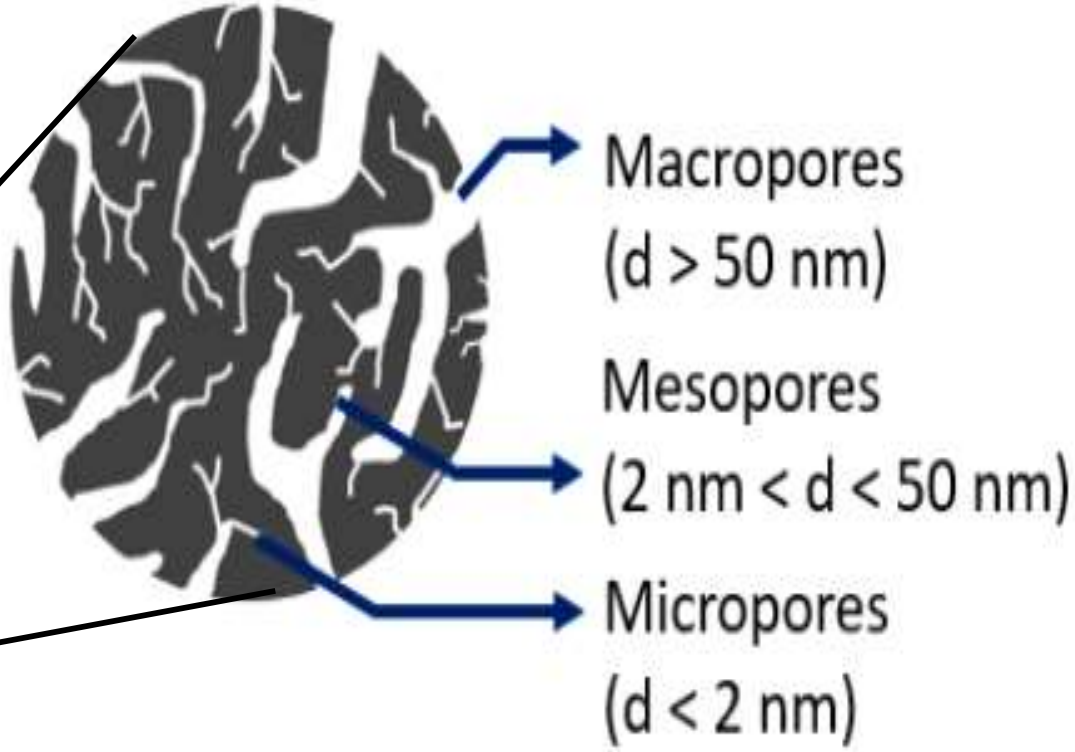
GAC

PAC

Expanded view of Internal Structure of Activated Carbon

Granular Activated Carbon

Granular Activated Carbon Particle



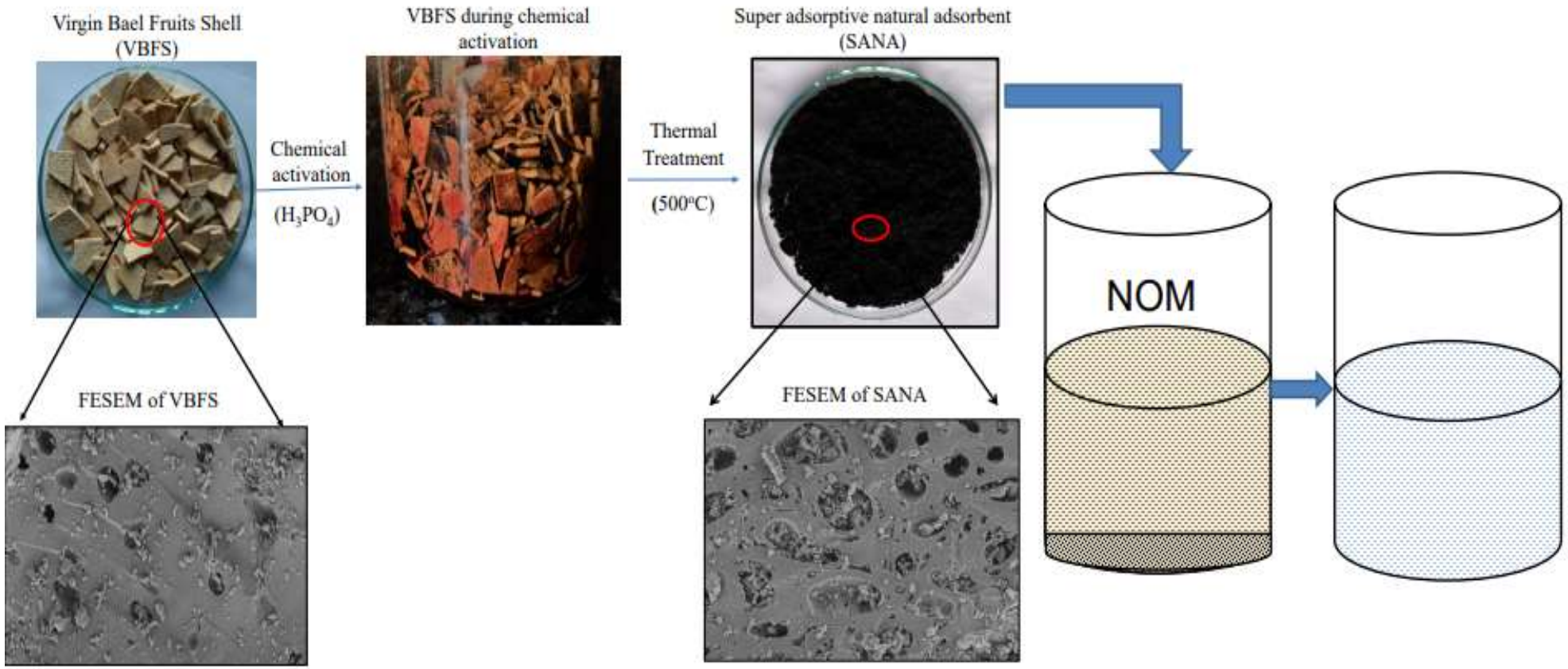
Comparison of granular and powdered activated carbon

Parameter	Unit	GAC	PAC
Particle size (dia)	mm	>0.1	<0.074
Total surface area	m^2/g	700-1300	800-1800
Bulk density	Kg/m^3	400-500	360-740
Wet Particle density	Kg/L	1.0-1.5	1.3-1.4
Particle size range	mm (µm)	0.1-2.36	(5-50)
Effective size	mm	0.6-0.9	NA
Uniformity coefficient	UC	≤1.9	NA
Mean pore radius	$\dot{A}$	16-30	20-40
Ash	%	≤8	≤6
Moisture as packed	%	2-8	3-10

NPTEL
JAN 2024

Synthesis of Natural Adsorbent Using Chemical Activation Process

Synthesis and evaluation of Super adsorptive natural adsorbent (SANA) derived from non-usable Bael fruits shell (BFS) for drinking water treatment – A corridor for THMs obliteration

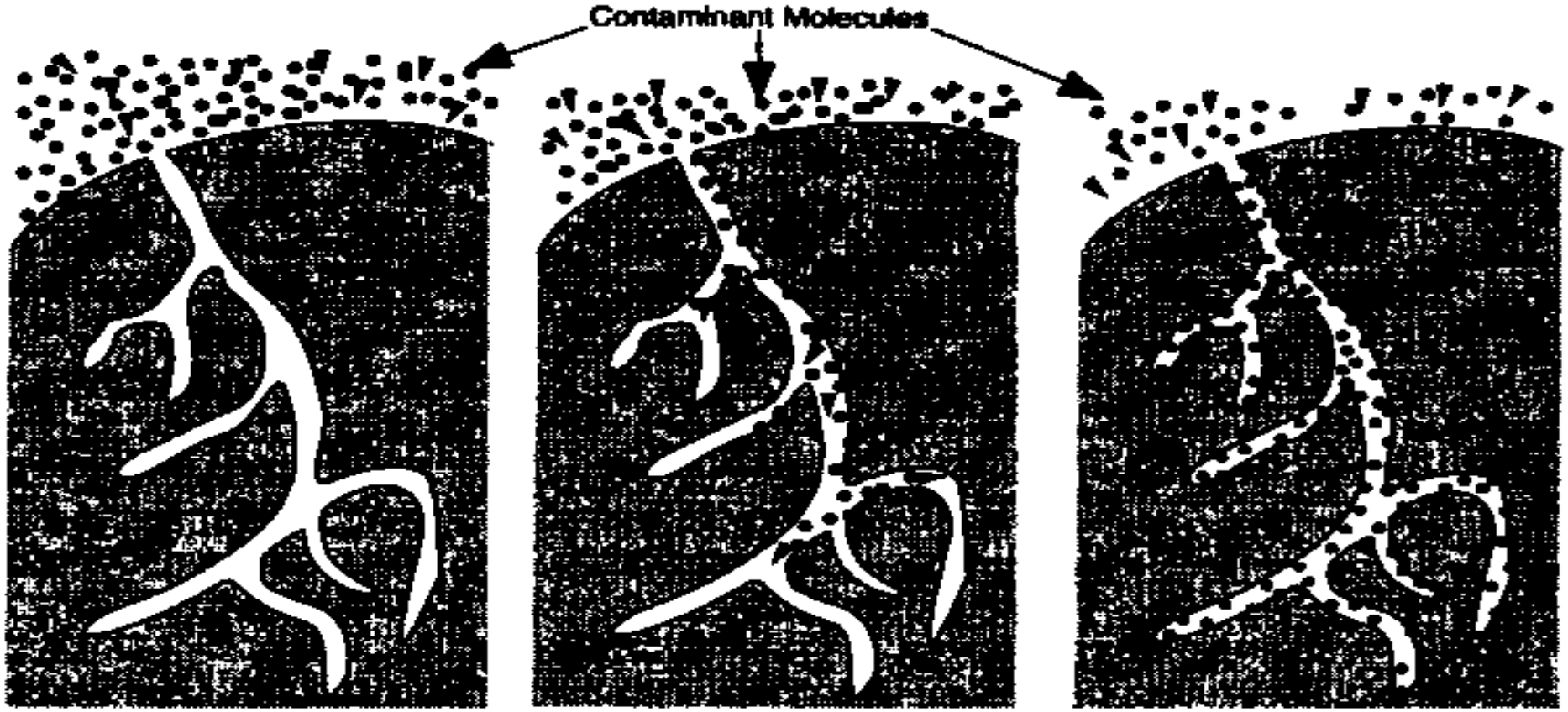


Mechanism of Adsorption

Step-1: Diffusion to Adsorbent Surface

Step-2: Migration into Pores of Adsorbent

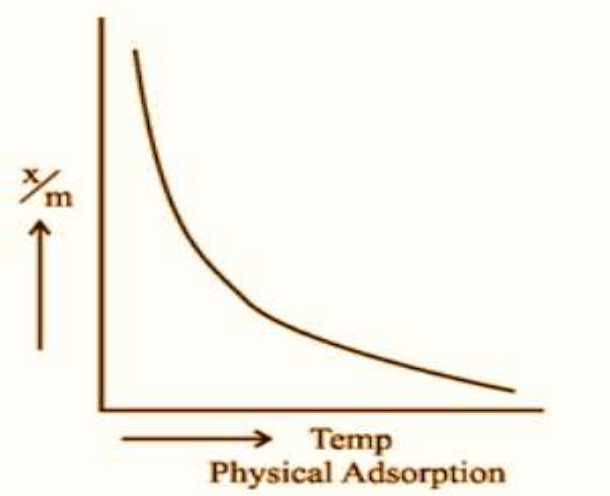
Step-3: Monolayer Buildup of Adsorbate



Factors affecting Adsorption

❖ Nature of Adsorbate/Gas

- Easily liquefiable gases **(NH_3, HCl, Cl_2, SO_2)** are readily adsorbed than permanent gases **(O_2, N_2, H_2).**
- Adsorption **α** (1/Solubility of adsorbate**)**

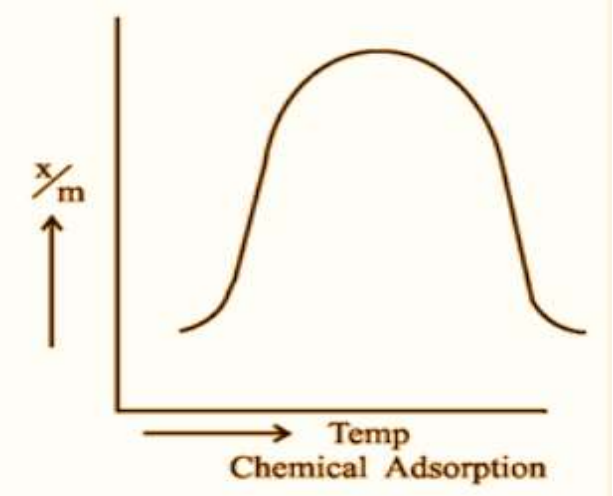


❖ Nature of Adsorbent and Surface area:

- Adsorbent is selective and specific in chemisorption
- Adsorption α Surface area of the adsorbent
- Presence of active groups on adsorbents (polar/activated) increases adsorption

❖ Temperature

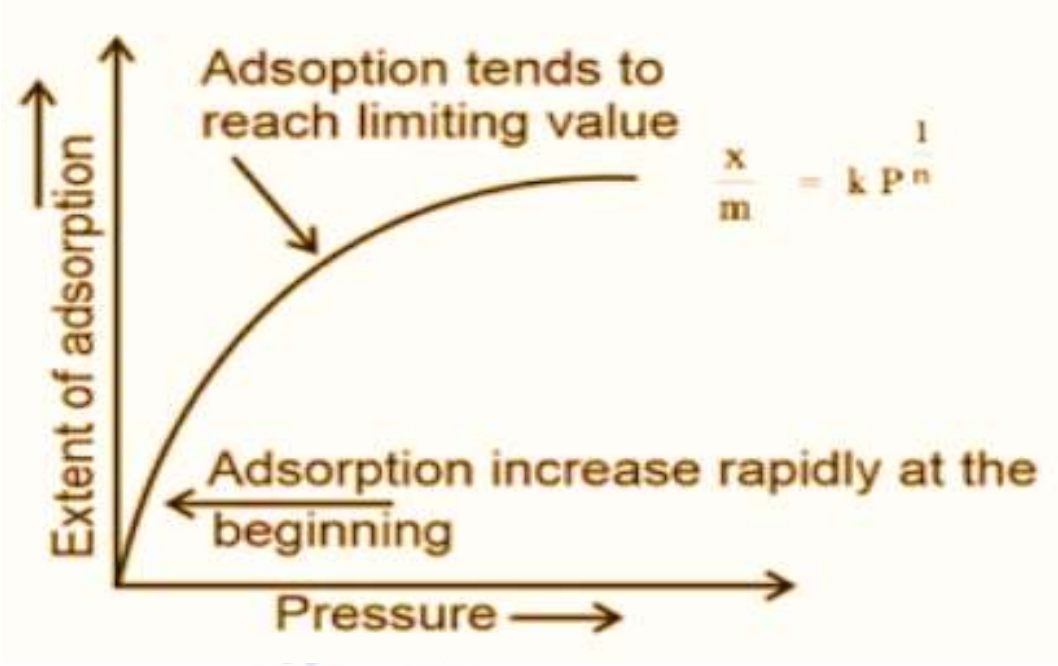
- Physisorption decreases when temperature increases
- Chemisorption increases with increase in temperature up to equilibrium limit then decreases with increase in temperature.



Factors affecting Adsorption

❖ Pressure

- Physisorption increases when pressure increases up to a limit and then the rate become constant
- The effect of pressure can be explained by Freundlich and Langmuir adsorption isotherm.
- Chemisorption is not affected by pressure



Application of Adsorption Process

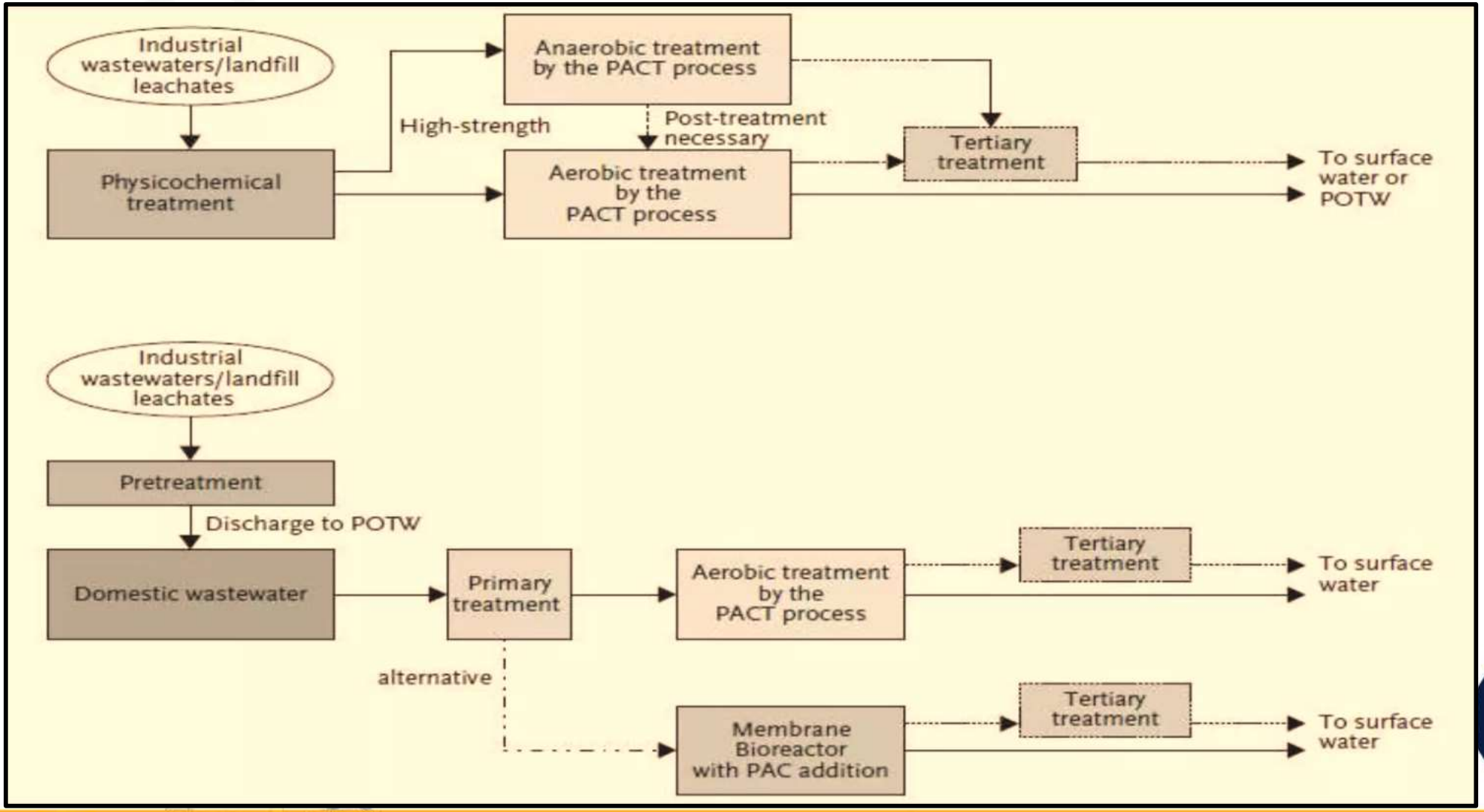
❖ **Water Treatment**

❖ **Wastewater Treatment (Secondary and Tertiary Treatment):** At current stage of development integrated systems are basically divided two categories based on the configuration of the biomass:

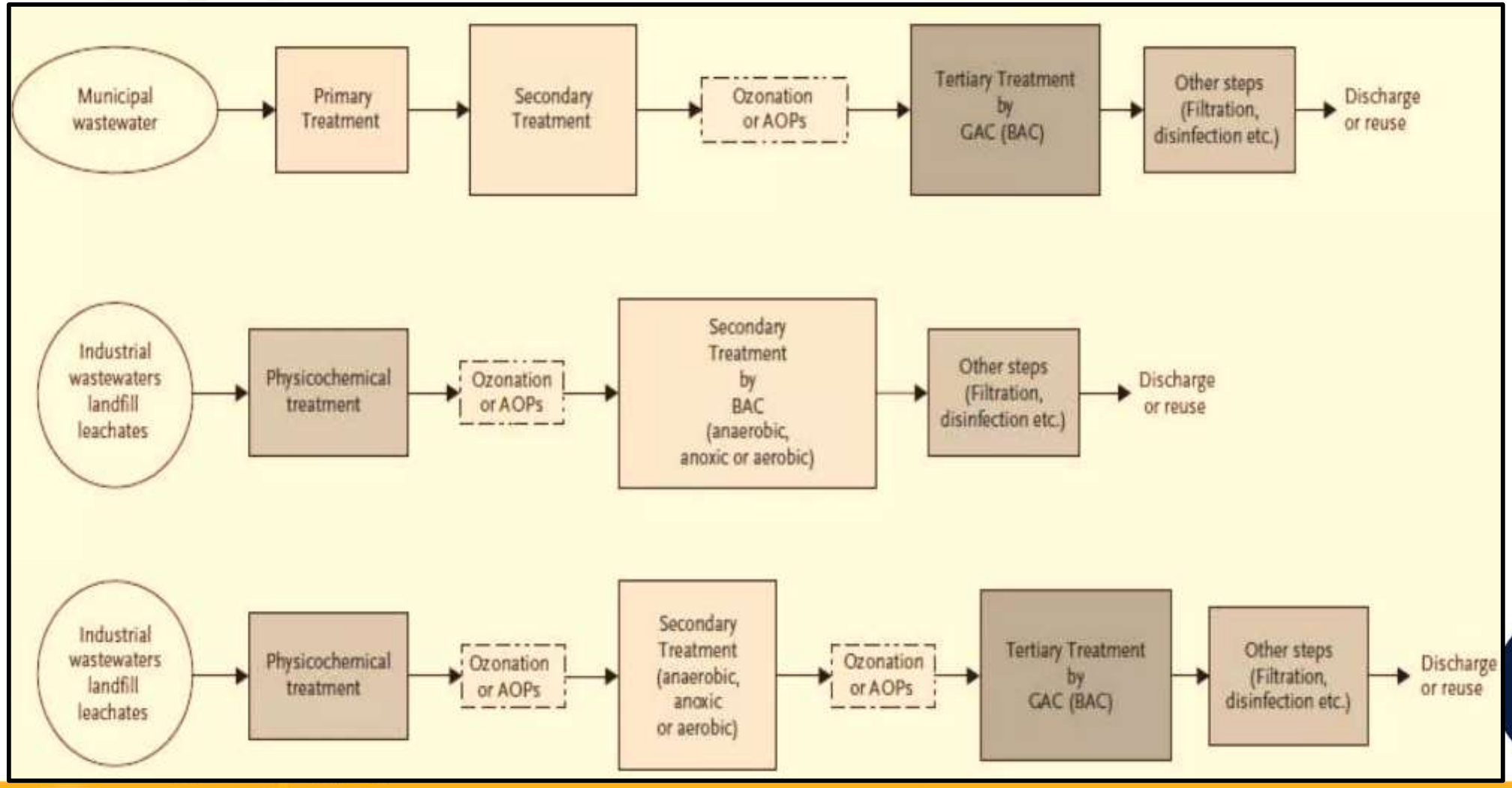
a) Suspended growth biological systems receiving activated carbon dosage: **Powdered Activated Carbon Treatment** (PACT) process and **PAC added membrane bioreactor** (PAC-MBR) process.

b) Attached-growth (biofilm) biological systems containing GAC media, namely **Biological Activated Carbon** (BAC) Process

Use of PAC in Secondary Treatment of Wastewater



Use of GAC in Secondary and Tertiary Treatment of Wastewater



Adsorption Isotherm

- Adsorption isotherm is a **mathematical model or functional expression** to **depict the distribution of solute between the solid and liquid phase** at equilibrium at constant fixed temperature.

Or

- The relation between the **quantity of adsorbate that can be adsorbed per unit weight of adsorbent**, at a constant temperature, is called an adsorption isotherm.

- The absorbent phase concentration after equilibrium is computed using following equation:

$$q_e = \frac{(C_0 - C_e) \times V}{m}$$

Where, q_e = adsorbent (i.e., solid) phase concentration after equilibrium, mg adsorbate/g adsorbent

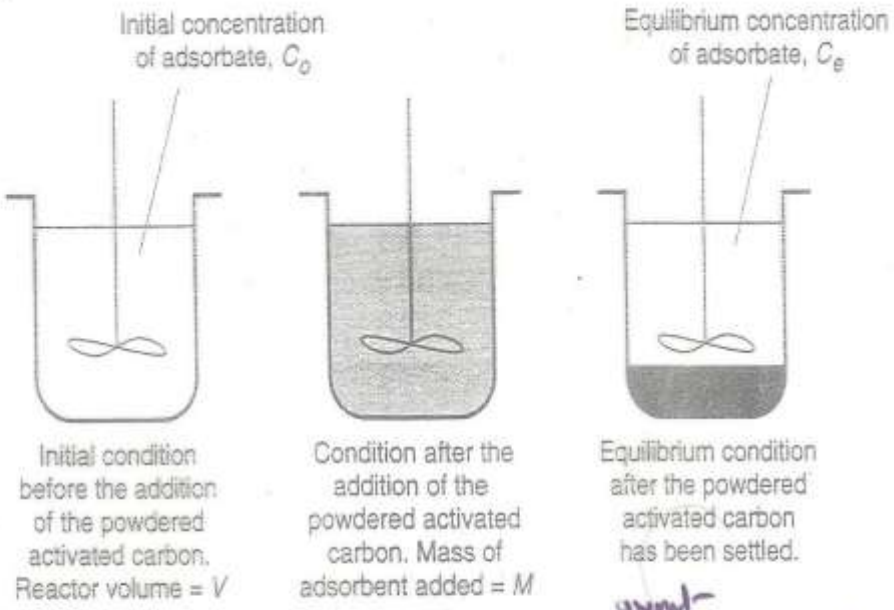
C_0 = initial concentration of adsorbate (mg/L)

C_e = final equilibrium concentration of adsorbate after absorption has occurred (mg/L)

V = volume of liquid in the reactor (L)

m = mass of adsorbent (g)

Development of Adsorption Isotherms



q_e = Mass of adsorbate / ~~Volume~~ wt of adsorbent mg/mg

M = Mass of adsorbent added mg

Mass balance :- Amount of adsorbate adsorbed = Initial amount of adsorbate in solution − Final amount of adsorbate in solution

$$q_e M = C_o V - C_e V$$

$$q_e = \frac{V}{M}(C_o - C_e)$$

Development of Adsorption Isotherm

- Batch Experiment is done by exposing a given amount of absorbate in a fixed volume of liquid to varying amounts of activated carbon.
- Typically, fixed volume of water (100-250 mL) is taken in more than ten closed cylinders
- Then different doses of adsorbent is added in different cylinders and shaken in shakers.
- The minimum time allowed for the samples to achieve adsorption equilibrium is seven days (for PAC).
- At the end of the test period, the samples is taken and amount of absorbate remaining in solution is measured.
- The analysis result is then plotted to get the adsorption isotherm.

Development of Adsorption Isotherm

❖ Mass Balance

$$Amount\ of\ Adsorbate\ Adsorbed$$
$$= Initial\ amount\ of\ adsorbate\ -\ Initial\ amount\ of\ adsorbate$$
$$in\ the\ solution \qquad\qquad in\ the\ solution$$

$$q_e \times M = C_0 \times V - C_e \times V \ ;\ \boldsymbol{q_e = (C_0 - C_e) \times \frac{V}{M}}$$

$\boldsymbol{q_e}$ = adsorbent (i.e., solid) phase concentration after equilibrium, mg adsorbate/g adsorbent

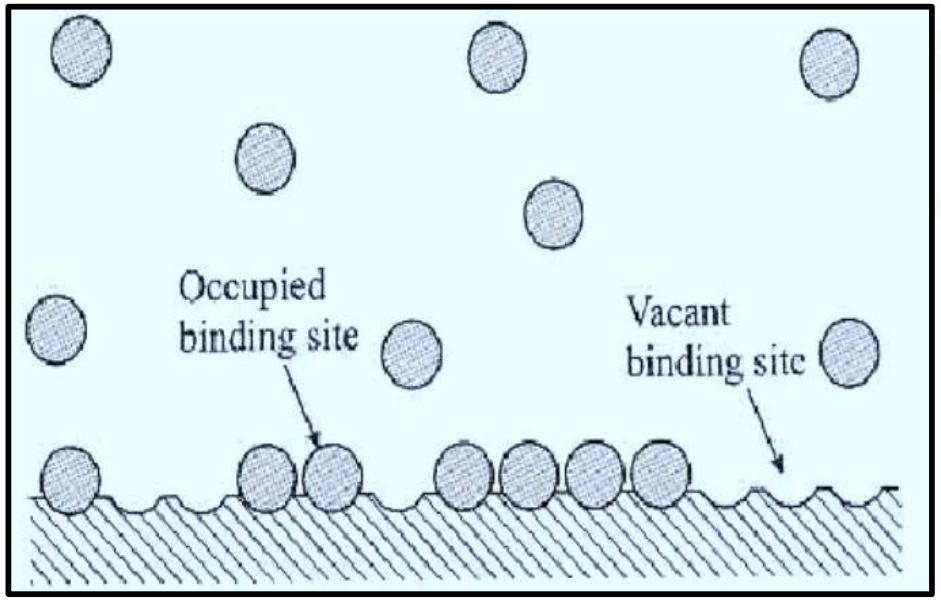
$\boldsymbol{C_0}$ = initial concentration of adsorbate (mg/L)

$\boldsymbol{C_e}$ = final equilibrium concentration of adsorbate after absorption has occurred (mg/L)

Types of Adsorption Isotherm

There are two different types of adsorption isotherms:

1. Langmuir Isotherm
2. Freundlich Isotherms



Langmuir Isotherm

1. Langmuir Isotherm

Assumptions made in the derivation of the Langmuir model are:

- Adsorption is a reversible process
- The adsorbed layer is made up of a single layer of molecules
- The adsorbed molecules do not move on the surface of the adsorbent.
- However, they can be desorbed back to the solution.
- The enthalpy of adsorption is the same for all

Derivation of Langmuir Isotherm

- In the Langmuir model (1918) adsorption is assumed to be a dynamic process.
- At equilibrium the number of molecules being adsorbed will be equal to the number of molecules leaving the adsorbed state, i.e., given by

$$Rate\ of\ adsorption \propto (Conc.\ in\ solution) \times (Available\ area\ for\ adsorption)$$

$$\propto C \times \left(A_{total} - A_{occupied}\right)$$

$$\propto C \times (K_1' \times q_m - K_1' \times q)$$

$$= K_1 C\ (q_m - \mathrm{q}) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(i)}$$

Similarly,

$$Rate\ of\ desorption \propto (Amount\ already\ adsorbed)$$

$$\propto A_{occupied}$$

$$\propto K_1' \times q = K_2 \times q \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(ii)}$$

At equilibrium,

$$\boldsymbol{Rate\ of\ adsorption = Rate\ of\ desorption}$$

Derivation of Langmuir Isotherm contd…….

At equilibrium,

$$\boldsymbol{Rate\ of\ adsorption\ =\ Rate\ of\ desorption}$$

$$K_1 C\,(q_m - \mathrm{q}) = K_2 \times q \quad \text{(from eqn i and ii)}$$

The previous eqn can be rewritten as:

$$q = \frac{q_m \times K_1 \times C}{K_2 + K_1 \times C} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \text{(iii)}$$

Substitute $K_A = \frac{K_1}{K_2}$ in eqn (iii), we will get the final expression for the **Langmuir Isotherm:**

$$\boxed{\boldsymbol{q = \frac{q_m K_A C}{1 + K_A C}}}$$

Where,

$\boldsymbol{q_m}$ **=** maximum adsorption capacity (mg/g)

$\boldsymbol{K_A}$ **=** Langmuir isotherm constant (function of enthalpy of adsorption and temperature)

Determination of Parameters in Langmuir Isotherm

The Langmuir isotherm can also be rearranged to give:

$$\frac{1}{q} = \frac{1}{q_m} + \left(\frac{1}{K_A q_m}\right)\frac{1}{C}$$

By plotting $\frac{1}{q}$ vs $\frac{1}{C}$ which will give:

Slope = $\boldsymbol{\frac{1}{K_A q_m}}$ **and** **Intercept =** $\boldsymbol{\frac{1}{q_m}}$

———————— × ————————

Freundlich Isotherms

❖ The Freundlich equation can be mathematically represented by:

$$\frac{x}{m} = K_f C_e^{1/n}$$

Where,

$\frac{x}{m}$ = mass of adsorbate adsorbed per unit mass of adsorbent (mg adsorbate/g activated carbon)

K_f = Freundlich capacity factor, (mg adsorbate/g activated carbon)(L water/ mg adsorbate)1/n

C_e = equilibrium concentration of adsorbate in solution after adsorption (mg/L)

$1/n$ = Freundlich intensity parameter

JAN 2024

Determination of Parameters in Freundlich Isotherm

The Langmuir isotherm can also be rearranged to give:

$$\log \frac{x}{m} = \log K_f + \frac{1}{n}\, logC_e$$

By plotting $\log \frac{x}{m}$ vs $logC_e$ which will give:

Slope = $\frac{1}{n}$ and Intercept = $\log K_f$

———————— × ————————

Classification of organics based on adsorption characteristics

Readily adsorbed organics

- Aromatic solvents
- Benzene
- Toluene
- Nitrobenzenes
- Chlorinated aromatics
- PCBs
- Chlorophenols
- Polynuclear aromatics
- Acenaphthene
- Benzopyrenes
- Pesticides and herbicides
- DDT
- Aldrin
- Chlordane
- Atrazine
- Chlorinated non aromatics
- Carbon tetrachloride
- Chloroalkyl ethers
- Trichloroethene
- Chloroform
- Bromoform
- High-MW hydrocarbons
- Dyes
- Gasoline
- Amines
- Humics

Poorly adsorbed organics

Low-MW ketones, acids, and aldehydes
Sugars and starches
Very-high-MW or colloidal organics
Low-MW aliphatics

Freundlich Isotherm Constants

Compound	pH	K,(mg/g)(L/mg)'/n	1/n
Benzene	5.3	1.0	1.6-2.9
Bromoform	5.3	19.6	0.52
Carbon tetrachloride	5.3	11	0.83
Chlorobenzene	7.4	91	0.99
Chloroethane	5.3	0.59	0.95
Chloroform	5.3	2.6	0.73
DDT	5.3	322	0.50
Dibromochloromethane	5.3	4.8	0.34
Dichlorobromomethane	5.3	7.9	0.61
1 ,2-Dichloroethane	5.3	3.6	0.83
Ethylbenzene	7.3	53	0.79
Heptachlor	5.3	1,220	0.95
Hexachloroethane	5.3	96.5	0.38
Methylene chloride	5.3	1.3	1.16
N-Dimethyln itrosamine	na	6.8 X 10-5	6.60
N-N itrosodi-n-propylamine	na	24	0.26
N-Nitrosodiphenylamine	3-9	220	0.37
PCB	5.3	14,100	1.03
PCB 1221	5.3	242	0.70
PCB 1232	5.3	630	0.73
Phenol	3-9	21	0.54
Tetrachloroethylene	5.3	51	0.56
Toluene	5.3	26.1	0.44
1,1, I-Trichloroethane	5.3	2-2.48	0.34
Trichloroethylene	5.3	28	0.62

Adapted from Dobbs and Cohen (1980) and LaGrega et al. (2001).

THANK YOU

NPTEL
NPTEL
JAN 2024